

CSC-355 Homework #5

first and follow sets

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This homework is an opportunity for you to practice computing first and follow sets for LL(1) grammars.

Problem 1: Generating a parse table for L

Answer questions (a)-(d) regarding the following LL(1) grammar.

- (1) $S \rightarrow A B$
- (2) $A \rightarrow \text{"a"} A$
- (3) $A \rightarrow \epsilon$
- (4) $B \rightarrow \text{"b"} B$
- (5) $B \rightarrow \epsilon$

- (a) Identify which, if any symbols are nullable.
- (b) Identify the first sets for each symbol.
- (c) Identify the follow sets for each symbol.
- (d) Generate the parse table for the list-manipulation language.

Problem 2: Generating a parse table for L

If we changed rule (2) to be: $A \rightarrow \text{"b"} A$, would the grammar from Problem 1 still be LL(1)? Why or why not?

Problem 3: Generating a parse table for list-manipulations

Answer questions (a)-(d) regarding the following LL(1) grammar.

- (1) $\text{Start} \rightarrow \text{Listhead ExprTail}$
- (2) $\text{Exprtail} \rightarrow \epsilon$
- (3) $\text{Exprtail} \rightarrow \text{"++"} \text{Start}$
- (4) $\text{Listhead} \rightarrow \text{Elem Listtail}$
- (5) $\text{Listhead} \rightarrow \text{"["}$
- (6) $\text{Listtail} \rightarrow \text{": ":"} \text{Listhead}$
- (7) $\text{Elem} \rightarrow \text{"a"}$

- (a) Identify which, if any symbols are nullable.
- (b) Identify the first sets for each symbol.
- (c) Identify the follow sets for each symbol.